

CI · 878 definitions

Unit / component / integration · GitHub Actions

Learning & model integrity 78

Data splits · aggregation math · model bindings

Round control & recovery 117

Local timer · input closure · quorum · publication

Identity & secure transport 240

Admission · quotes · keys · TLS · authorization

Inference & agent workflow 49

MCP tools · job ownership · TEE/ZK workflow

Receipts & provenance 64

AIR / Sello / SCITT · signatures · data provenance

Runtime control & telemetry 107

Configuration · upload gates · worker telemetry

Deployment & packaging 223

Image policy · revision order · Docker / Terraform

External · 8 modes + audit

SMEW → SMA → vita-fl-td · deployed VITA-FL in Phala

Settings · configuration

Save/read parameters; reject invalid deadlines

Lifecycle · normal / inference

Complete training; use and verify the finalized model

Collection · early / partial / zero

Close early, publish partial inputs, or recover from zero

Recovery · aggregator outage

Stop CVM; 1/3 → 2/3 → 3/3 votes; elect a replacement

Security · access and contract guards

16 checks: access, contracts + positive inference

Evidence · published receipt audit

17 checks: SCITT, AIR/TDX, sessions and Sello

Phala training profile: 6 workers · 5 successful rounds · client limit 5 · 45 s deadline. Counts ≠ passed runs.

One system, two complementary test boundaries



EVALUATION · CI COVERAGE AND EXTERNAL TEST CONFIGURATION

878 CI definitions + 8 external scenario modes and a receipt audit

TESTED AREA	CI #	CI · UNIT / COMPONENT / INTEGRATION	EXTERNAL · DEPLOYED VITA-FL
Learning & model integrity	78	Data partitions; aggregation math; model-bundle authentication	normal: 5 completed training rounds All six workers participate
Round control & recovery	117	Local timer; closure; quorum; atomic publication and round state	early / partial / zero; recovery Deadline, 3-vote quorum, resumed rounds
Identity & secure transport	240	Admission policies; key binding; RA-TLS/mTLS and access control	security: reject missing/forged auth Reject invalid coordination calls
Inference & agent workflow	49	Tool API; job ownership; TEE and ZK inference lifecycle	inference: finalized-model binding Deployed agent and actual inference
Receipts & provenance	64	AIR/Sello checks; log bindings; signed training-data provenance	17-check published-evidence audit SCITT · AIR/TDX · sessions · Sello
Runtime control & telemetry	107	Configuration validation; gate control; signed-event metrics	configuration: exact settings readback Invalid deadlines leave state unchanged
Deployment & packaging	223	Images and deployment order; Docker contexts; offline Terraform	No external redeployment scenario Run pins: image, chain and contracts

CI: GitHub Actions / Ubuntu, push + PR + manual. External: SMEW / SMA / TD → Phala.

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GitHub Actions · Ubuntu · push / PR / manual	

8 external scenario modes

SMEW / SMA / vita-fl-td → Phala

SCENARIO	CONDITION → EXPECTED RESPONSE
configuration	Readback matches; invalid deadline rejected
normal	5 completed rounds; final model verified
early	Accept 5/5 → close before the deadline
partial	Accept 2/5 → publish those 2 after timeout
zero	Accept 0/5 → no publication; quorum recovery
recovery	Stop aggregator → quorum → new aggregator
inference	Verify inference using the finalized model
security	16 checks: access, contracts + positive control

Additional audit: 17 checks of published SCITT / AIR / TDX / Sello evidence

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